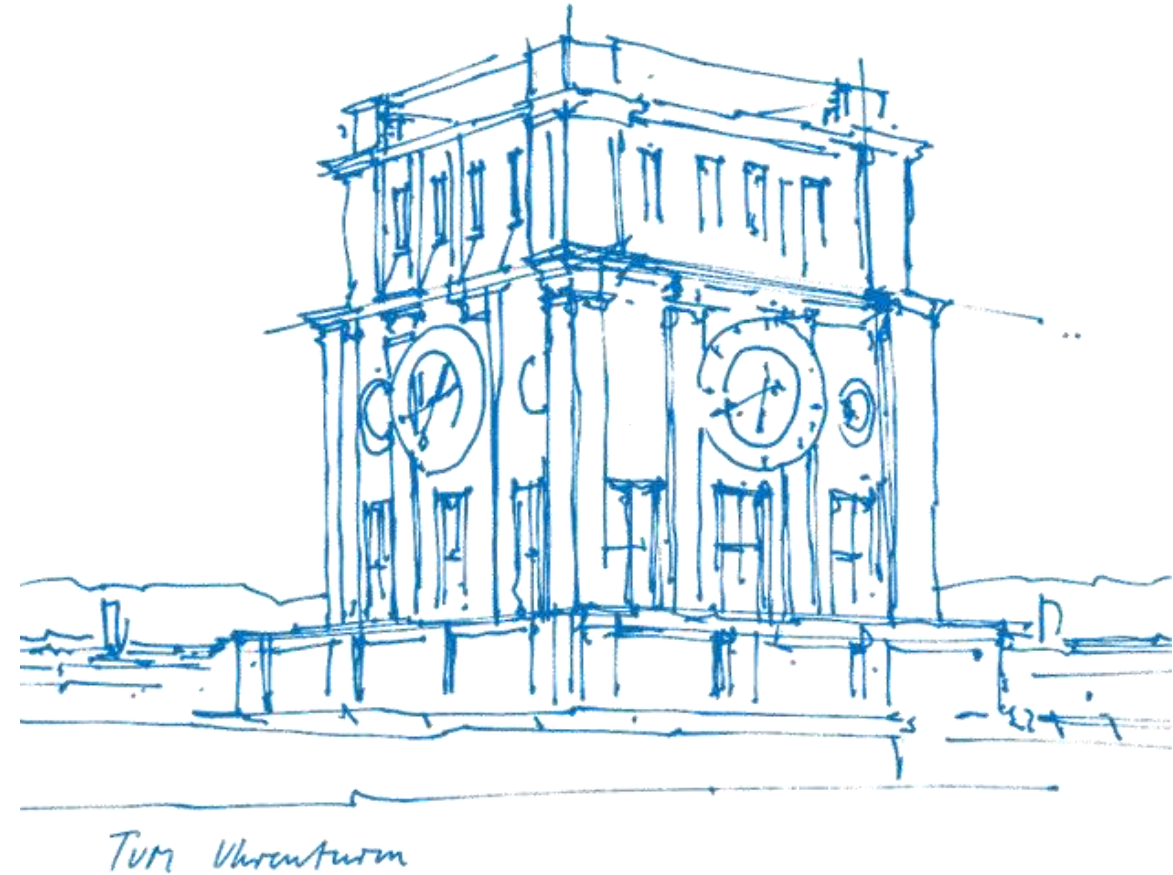


Tutorial 00

Set Theory and Introduction of Probability

Xujie Zhao

20. April / 21. April 2026



- **Lecture:** 2.5h per week at 16:00-18:30 Wed, held by Prof. Niki Kilbertus in room MW2001
- **Tutorial:** In-Person, 1.5h by two time periods per week
 - 12:15-13:45 Mon, 00.08.053 (A31)
 - 14:15-15:45 Tue, 00.08.053 (B42)
- The slides are not guaranteed to be accurate! (Written by myself)
- **Language during the tutorial:** Mainly English, Deutsch may be used if needed, Chinese might be used during calculation (that would be an exception and will be repeated in English)
- **Exam:** August 5, 2026, 08:00-10:00, Cheatsheet allowed, Calculator banned
- **Materials of this tutorial:** Please scan the QR-Code 
- **Contact:** xujie.zhao@tum.de



Overview (referring to last year's lecture)

- **Discrete Probability**

- Probability Space, Event, Random Variable
- Special Distributions
- Markov's and Chebyshev's Inequalities

- **Continuous Probability**

- Normal Distribution, Exponential Distribution
- Central Limit Theorem

- **Statistics**

- Estimators
- Confidence Intervals
- Hypothesis Testing

△ Derivative and Integral required!
(Analysis für Informatiker MA0902)

Personal advice...

- PART I. Recap of the Lecture
- PART II. Exercises from the Exercise Sheet
- PART III. Additional Exercises (with solution)

I. Recap

- **Set Theory and Probability**

- Sets and basic operations
- Samples (sample space), Events, Event Systems
- Probability of simple events

- **Conditional Probability**

- Definition

II. Exercise Sheet



- Exercise 1.**
- (i) You roll two fair six-sided dice. What is the probability that the sum is 7?
 - (ii) What is the probability that the sum is at least 10?
 - (iii) What is the probability that at least one die shows a 6?

II. Exercise Sheet

Exercise 2. An urn contains 3 red balls and 2 blue balls. You draw two balls uniformly at random without replacement.

- (i) What is the probability that both balls are red?
- (ii) What is the probability that you draw exactly one red ball?
- (iii) What is the probability that the second ball is blue?
- (iv) Given that the second ball is blue, what is the probability that the first ball is red?
- (v) Given that at least one of the two balls is blue, what is the probability that exactly one of them is blue?

II. Exercise Sheet

Exercise 3. A 4-digit PIN is chosen uniformly at random from $\{0000, 0001, \dots, 9999\}$.

- (i) What is the probability that all four digits are distinct?
- (ii) What is the probability that at least one digit is repeated?
- (iii) What is the probability that there is exactly one pair, meaning one digit appears exactly twice and the other two digits are different from each other and from the repeated digit?
- (iv) What is the probability that the PIN is a palindrome, meaning it reads the same forwards and backwards?

II. Exercise Sheet



Exercise 4. An automatic bug detector is used on student submissions. Suppose that 2% of all submissions actually contain a certain bug. The detector flags 90% of buggy submissions, but it also flags 5% of bug-free submissions.

- (i) Imagine 10 000 submissions. How many buggy submissions would be flagged?
- (ii) How many bug-free submissions would be flagged?
- (iii) If a randomly chosen submission was flagged, what is the probability that it actually contains the bug?

II. Exercise Sheet

Exercise 5. A standard deck has 52 cards, including 4 aces.

- (i) You are dealt two cards and look at the first one. It is an ace. What is the probability that the second card is also an ace?
- (ii) Now consider a different situation: You are again dealt two cards, but this time you do not look at them. Instead, someone truthfully tells you that at least one of your two cards is an ace. Is the probability that both cards are aces the same as in (i)?

II. Exercise Sheet



Exercise 6. If you choose an answer to this question at random, what is the probability that you will be correct?

A 25%

B 50%

C 0%

D 25%

III. Additional Exercises

Are these two statements **true or **false**?**

1. An event that is impossible to happen has the probability of 0.
2. An event that has the probability of 0 is impossible to happen.

Materials

